



023

**University of Colombo, Sri Lanka***University of Colombo School of Computing***Bachelor of Science in Computer Science****Bachelor of Science Honours in Computer Science****Bachelor of Science Honours in Software Engineering**

Third Year Examination — Semester I - 2022

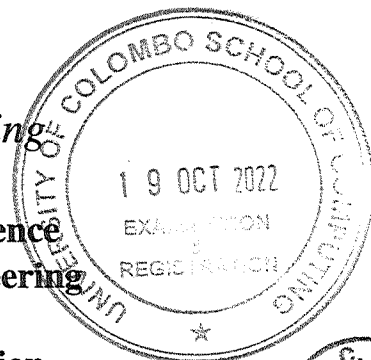
IS 3109 /SCS3210 — Systems & Network Administration

Two (2) Hours

Answer All Questions

Number of Pages = 11

Number of Questions = 4

**To be completed by the candidate**

Index Number

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Important Instructions to candidates

- The duration of the paper is 2 Hours.
- The medium of instructions and questions is English.
- You shall write your answers in **English**.
- This paper has 4 questions on 11 pages.
- Answer **all 4** questions.
- **Write your answers on and only on the space provided** on this question paper.
- Write your index number on each and every page of the question paper.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries and mobile phones, are **not allowed**.
- Non-programmable Calculators may be used.

To be completed by the examiners

1	
2	
3	
4	
Total	

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1. (a). “The number of virtual hosting sites is limited by the number of IP addresses the web server has been configured with”

Is the above statement **true** for **name based virtual hosting** in Apache web server? Explain your answer.

[5 marks]

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- (b). You are being asked to deploy an Apache Tomcat server which uses a MySQL database. Write the main steps that you need to perform in deploying the solution. Step 01 is provided as an example.

Step 01: Install Apache Tomcat and test if it is working.

[5 marks]

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(c). List three (03) examples for permissive licenses.

[3 marks]

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(d). Following is the answer received for `dig www.google.lk` command. Describe what is meant by each component of the answer received.

`www.google.lk` 213 IN A 172.217.194.94

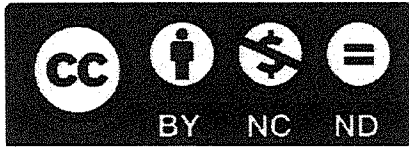
[5 marks]

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(e). Write the terms and conditions of the creative commons license depicted in the below image.



[5 marks]

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(f). List two (02) best practices a systems and network administrator shall practice.

[2 marks]

i. _____

ii. _____

Index Number

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2. (a). Documentation and backup is considered as most important practices of systems and network administration. Logs can be considered as a document containing the information about sequence of events happened with respect to a particular service.

i. Write **two (02)** benefits of having log rotation implemented on a particular log file.

[2 marks]

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ii. Explain the following log rotation configuration.

```
/var/log/sss/*.log {  
    weekly  
    missingok  
    notifempty  
    sharedscripts  
    rotate 2  
    compress  
    postrotate  
        /bin/kill -HUP `cat /var/run/sss.pid 2>/dev/null`  
    endscript
```

[10 marks]

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- iii. Explain how a remote incremental backup script/system can be implemented using linux hard links, ssh and rsync utility .
Hint: Use linux commands to explain.

[8 marks]

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iv. Explain the schedule of the linux cron configuration given below.

`*/30 4,16 * * mon,wed,fri /usr/bin/backup_script`

[5 marks]

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- 3. (a).** Discuss why installing the desktop versions of operating systems on server machines is not recommended.

[5 marks]

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- (b). Manjula has been appointed as the coordinator of the Network Operating Center of a particular organization. He proposed installing Red Hat Linux on newly purchased servers instead of CentOS.

- i. Sanjeewa, the CEO of the company, did not approve the proposal of Manjula. Discuss the reasons which influenced the decision of Sanjeewa.

[4 marks]

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- ii. Consider you as Manjula and construct the counterargument to prove what you have proposed is the best solution.

[6 marks]

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- (c). Write down the Linux commands to perform the following activities.

- i. Read the manual page of the command used to list down the content of a directory.

[1 marks]

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- ii. Create a new directory named “SCS3210” inside your home directory.

[2 marks]

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- iii. Create an empty file named “Questions” inside the “SCS3210” directory.

[2 marks]

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- iv. Make a copy of the “Questions” file as “Questions2” and rename it as “Questions.back”

[3 marks]

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- v. Delete the “Questions” file.

[2 marks]

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- [8 marks]**



- [3 marks]

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- [4 marks]

[illegible]

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- ii. What is the command you have to issue on the CentOS server machine installed in the DHCP service to check whether the service is up and running?

[3 marks]

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- iii. Since Indika needs to send some urgent emails, you were asked to assign an IP addresses manually to Indika's computer. Write down the Linux commands you have to run on Indika's computer to assign IP address 192.168.10.100/24 and the default gateway 192.168.10.254. (Assume that the operating system of Indika's computer is CentOS and aame of the network interface is eth0)

[7 marks]

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